UNIVERSALLY PARAMETRIZING THE OPEN-AND-CLOSED SUBSCHEMES OF A QUASI-FINITE MAP THAT ARE FINITE OVER THE BASE

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 22834-22845 of sources/stacks-more-morphisms/pinned-original-source.tex; proof span is lines 22847-22888, the complete author proof (40 body lines).

Editorial note. Navigation stubs: 4 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. The quasi-finite companion of the chapter's finite-case moduli-of-open-and-closed-subschemes lemma; the author's automatic-closedness sentence for open immersions between schemes finite over Y' is the only inline-implicit step and is a one-line standard fact.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Morphisms, lemma locally-quasi-finite-rel-dimension-0; the chapter's Noetherian-approximation-dimension-d and open-and-closed-in-finite lemmas (both previously collected); Algebra, lemma quasi-finite-open-integral-closure (quasi-finite morphisms are opens of finite ones after integral closure); standard idempotent/open-and-closed dictionary.

Source reference. Original lemma 00QB: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 22888–22888. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 05FJ: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 9950–9957. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0F3M: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 22791–22797. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0397: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 22888–22888. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Date: stacks-project:more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Lemma 0.1. *Let $X \to Y$ be a morphism of affine schemes which is quasi-finite and of finite presentation. There exists a morphism $Y_{univ} \to Y$ of finite presentation and an open subscheme $U_{univ} \subset Y_{univ} \times_Y X$ such that $U_{univ} \to Y_{univ}$ is finite with the following property: given any morphism $Y' \to Y$ of affine schemes and an open subscheme $U' \subset Y' \times_Y X$ such that $U' \to Y'$ is finite, there exists a morphism $Y' \to Y_{univ}$ such that the inverse image of U_{univ} is U'.*

Proof. Recall that a finite type morphism is quasi-finite if and only if it has relative dimension 0, see Morphisms, Lemma 0397. By Lemma 05FJ applied with $d = 0$ we reduce to the case where X and Y are Noetherian. We may choose an open immersion $X \to X'$ such that $X' \to Y$ is finite, see Algebra, Lemma 00QB. Note that if we have $Y' \to Y$ and U' as in (2), then

$$U' \to Y' \times_Y X \to Y' \times_Y X'$$

is open immersion between schemes finite over Y' and hence is closed as well. We conclude that U' corresponds to an idempotent in

$$\Gamma(Y', \mathcal{O}_{Y'}) \otimes_{\Gamma(Y, \mathcal{O}_Y)} \Gamma(X', \mathcal{O}_{X'})$$

whose corresponding open and closed subset is contained in the open $Y' \times_Y X$. Let $Y'_{univ} \to Y$ and idempotent

$$e'_{univ} \in \Gamma(Y_{univ}, \mathcal{O}_{Y_{univ}}) \otimes_{\Gamma(Y, \mathcal{O}_Y)} \Gamma(X', \mathcal{O}_{X'})$$

be the pair constructed in Lemma 0F3M for the ring map $\Gamma(Y, \mathcal{O}_Y) \to \Gamma(X', \mathcal{O}_{X'})$ (here we use that Y is Noetherian to see that X' is of finite presentation over Y). Let $U'_{univ} \subset Y'_{univ} \times_Y X'$ be the corresponding open and closed subscheme. Then we see that

$$U'_{univ} \setminus Y'_{univ} \times_Y X$$

is a closed subset of U'_{univ} and hence has closed image $T \subset Y'_{univ}$. If we set $Y_{univ} = Y'_{univ} \setminus T$ and U_{univ} the restriction of U'_{univ} to $Y_{univ} \times_Y X$, then we see that the lemma is true. □